PHL： 溯源超图启发式学习器（ Provenance-Hypergraph Heuristic Learner ）

在 PLMF-MDA 支持图上叠加策展溯源超图与 miRNA 旁信息关系，确定性计算 128 列启发式，接一个逻辑回归读头（约 130 个参数）。单模型、线性、无泄漏。

① 数据源

PLMF-MDA 训练正例

miRNA–药物关联矩阵（ MDR / MDS ），仅训练折的正例

iNcRD-HG 溯源记录

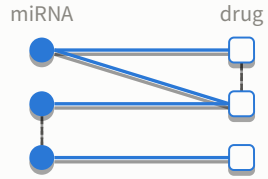
每条关联的 PMID ，以及报道它的细胞系 / 表型 / 疾病语境

miRNA 外部知识（标签无关）

HMDD v4 疾病谱 · 名称 / seed 家族 · 另一表型矩阵中的度

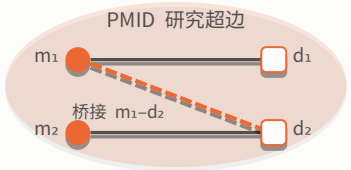
② 多关系超图（仅用 support 正例构建）

支持图 + kNN 内容图



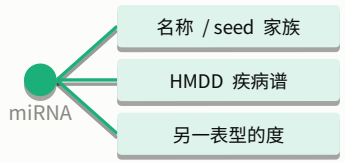
二部支持图：仅训练正例
kNN 内容图： miRNA / 药物相似度
实线 = 支持边；虚线 = kNN 边

溯源超图



研究超边：同一 PMID 报道的全部 (m, d)
语境超边：细胞系 / 表型 / 疾病 token
共研究图 $S = M \cdot M^T$

miRNA 旁信息关系



miRNA → 家族 / 疾病 / 跨表型度
标签无关，专门给零度、低度 miRNA 补信息

③ 启发式特征（ 128 列）

ctx (33) + diff (52)

度、共邻、 path3/5 、 SVD16 、属性支撑
PPR / Katz / path7 及行、列归一化
（原 iNcRD-HG 特征库）

prov (24)

PMID 桥接 $|P_m \cap P_d|$ 、规模加权、 Jaccard (6)
共研究传递 $(S_r A)$ 、 $(A S_d)$ (6)
语境 token 交集 (4) · 记录统计 (8)

mprior (19)

名称家族 (5) · seed 7-mer 家族 (5)
HMDD 疾病桥 $\Sigma H[m,dz] \cdot (H^T A)[dz,d]$ (7)
另一表型的度 xdeg (2)

④ 线性读头

无泄漏交叉拟合

训练行： K=10 内折， query 对绝不进入自己的 support
测试行：用全部训练正例作 support
所有启发式只读 support 正例

拼接 128 列



ctx+diff 85　prov 24　mprior 19

z = signed log1p + 标准化
$score(m, d) = w \cdot z + b$
逻辑回归， C = 0.3 ，约 130 个参数

score(m, d)

MDR · AUC / AUPR （论文 5 折， 4 seeds ）
0.9591 / 0.9565
+0.0062 / +0.0071 vs 基线 0.9529 / 0.9494 （ 4/4 seeds ）

MDS · AUC / AUPR （论文 5 折， 4 seeds ）
0.9607 / 0.9587
+0.0047 / +0.0051 vs 基线 0.9560 / 0.9536 （ 4/4 seeds ）

打乱对照（溯源与旁信息置换）
Δ ≈ 0
+0.0007 / +0.0011 AUC ，增益不是来自列数

增益落在哪里
稀疏 miRNA
支持度 1–7 的正例损失下降 8–15% ， 16+ 持平

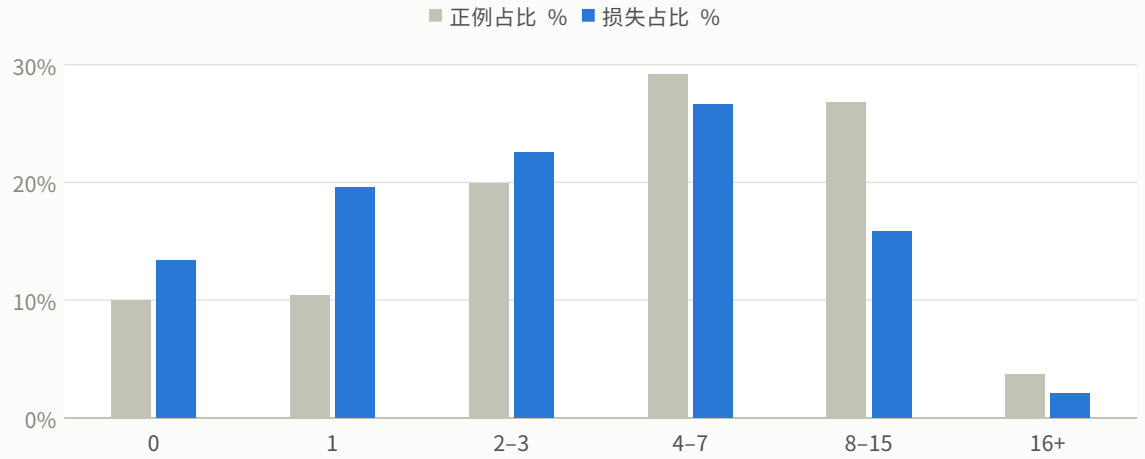
数值来自 REPORT_WARM_PHL.md §0 、 §4.1 、 §4.3 ；基线 = ctx+diff × LR （ K=10 ）；对 PLMF-MDA 官方复现领先 +0.033~0.035 AUC / +0.040~0.042 AUPR 。

诊断： warm 残差集中在稀疏 miRNA ， PHL 的增益也落在那里

seed 42 ，论文 5 折。损失 = 排在每个测试正例前面的负例比例（均值 = 1 − AUC ）；按 miRNA 在训练支持图中的度数分桶。

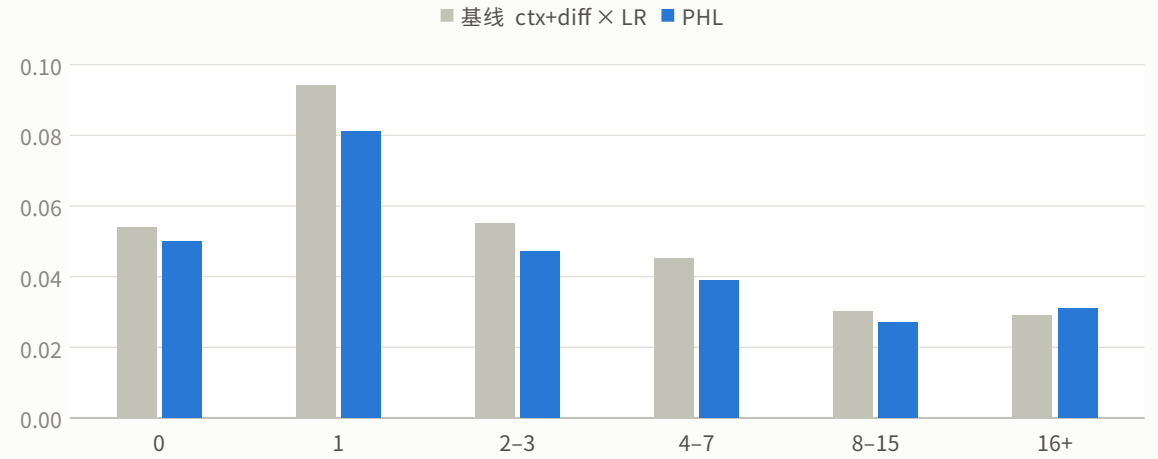
MDR · 正例占比 vs 损失占比

度 ⩽ 3： 正例 40% ，损失 55%



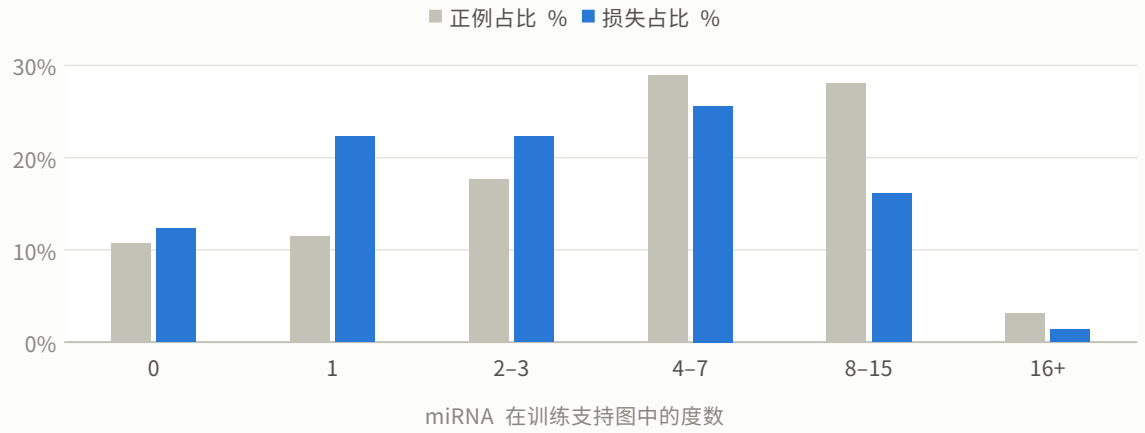
MDR · 正例侧单例均损：基线 → PHL

度 1–7 相对下降 13–15%； 16+ 持平



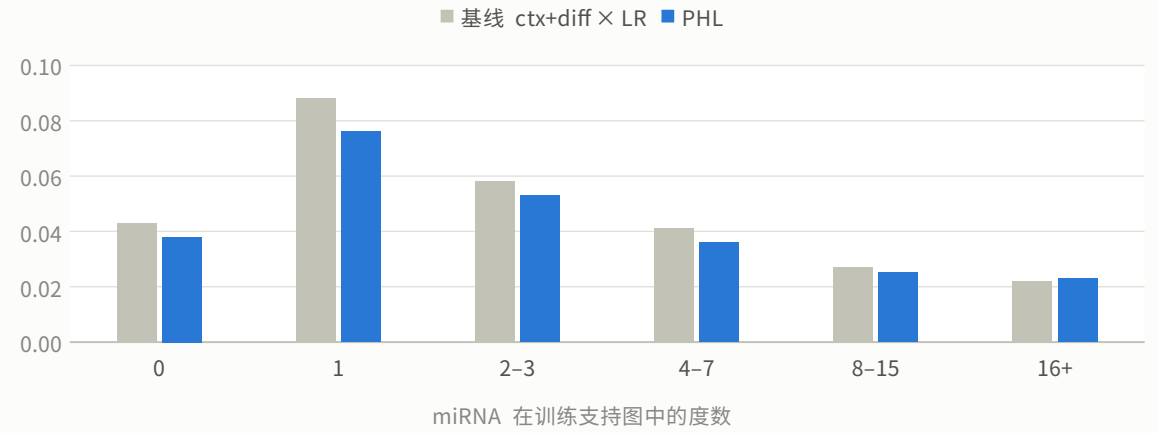
MDS · 正例占比 vs 损失占比

度 ⩽ 3： 正例 40% ，损失 57%



MDS · 正例侧单例均损：基线 → PHL

度 1–7 相对下降 8–14%； 16+ 持平



数值来自 REPORT_WARM_PHL.md §1.1 、 §4.3 （error_analysis.py）。

PHL: Provenance-Hypergraph Heuristic Learner — single-model architecture

Curated provenance hyperedges and miRNA side-information relations are stacked on the PLMF-MDA support graph; 128 deterministic heuristics feed one logistic-regression readout (~130 parameters). Single model, linear, leak-free.

① Data sources

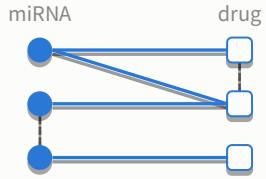
PLMF-MDA training positives
miRNA–drug association matrix (MDR / MDS), training-fold positives only

iNcRD-HG provenance records
PMID of every association plus the cell-line / phenotype / disease context it was reported in

External miRNA knowledge (label-free)
HMDD v4 disease spectrum · name / seed family · degree in the other phenotype matrix

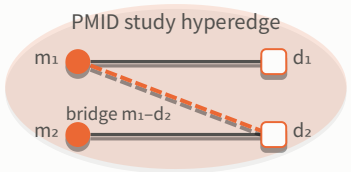
② Multi-relational hypergraph (support only)

Support graph + kNN content graph



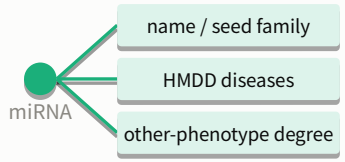
Bipartite support graph: training positives only
kNN content graph: miRNA / drug similarity
solid = support edge; dashed = kNN edge

Provenance hypergraph



Study hyperedge: all (m, d) reported by one PMID
Context hyperedge: cell-line / phenotype / disease tokens
Co-study graph $S = M \cdot M^T$

miRNA side-information relations



miRNA → family / disease / cross-phenotype degree
label-free; adds information for zero- and low-degree miRNAs

③ Heuristic features (128 cols)

ctx (33) + diff (52)
degree, common neighbours, path3/5, SVD16, attribute support
PPR / Katz / path7 with row / column normalisation
(original iNcRD-HG feature bank)

prov (24)
PMID bridging $|P_m \cap P_d|$, size-weighted, Jaccard (6)
co-study transfer ($S_r A$), ($A S_d$) (6)
context-token overlap (4) · record statistics (8)

mprior (19)
name family (5) · seed 7-mer family (5)
HMDD disease bridge $\Sigma H[m,dz] \cdot (H^T A)[dz,d]$ (7)
degree in the other phenotype, xdeg (2)

④ Linear readout

Leak-free cross-fitting
Training rows: K=10 inner folds; a query pair never enters its own support
Test rows: all training positives as support
Every heuristic reads support positives only

Concatenate 128 columns



ctx+diff 85 · prov 24 · mprior 19

z = signed log1p + standardise
score(m, d) = w · z + b
logistic regression, C = 0.3, ~130 parameters

score(m, d)

MDR · AUC / AUPR (paper 5-fold, 4 seeds)
0.9591 / 0.9565
+0.0062 / +0.0071 vs baseline 0.9529 / 0.9494 (4/4 seeds)

MDS · AUC / AUPR (paper 5-fold, 4 seeds)
0.9607 / 0.9587
+0.0047 / +0.0051 vs baseline 0.9560 / 0.9536 (4/4 seeds)

Shuffle control (provenance + side info permuted)
Δ ≈ 0
+0.0007 / +0.0011 AUC: the gain is not the extra columns

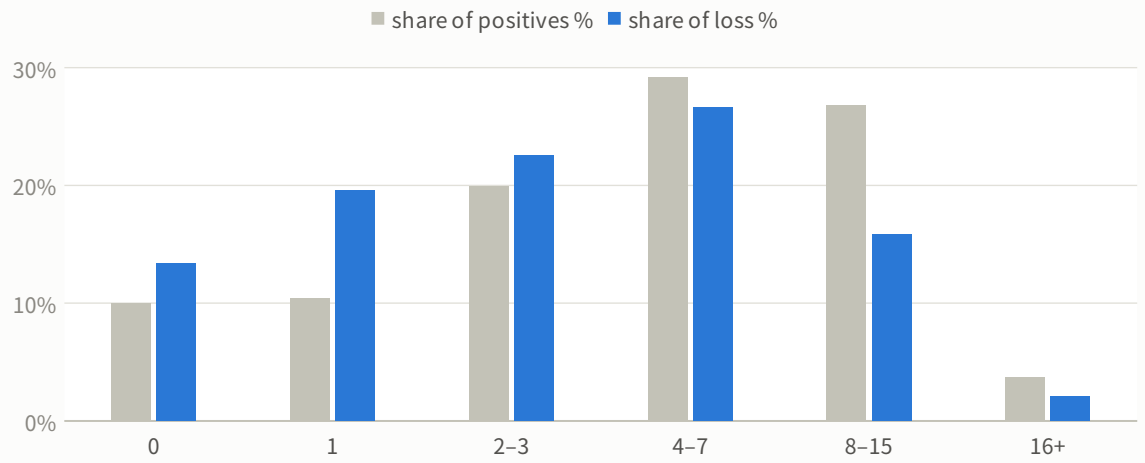
Where the gain lands
sparse miRNAs
loss on support-degree 1–7 positives drops 8–15%; 16+ flat

Numbers from REPORT_WARM_PHL.md § 0, § 4.1, § 4.3; baseline = ctx+diff × LR (K=10); +0.033–0.035 AUC / +0.040–0.042 AUPR over the PLMF-MDA official reproduction.

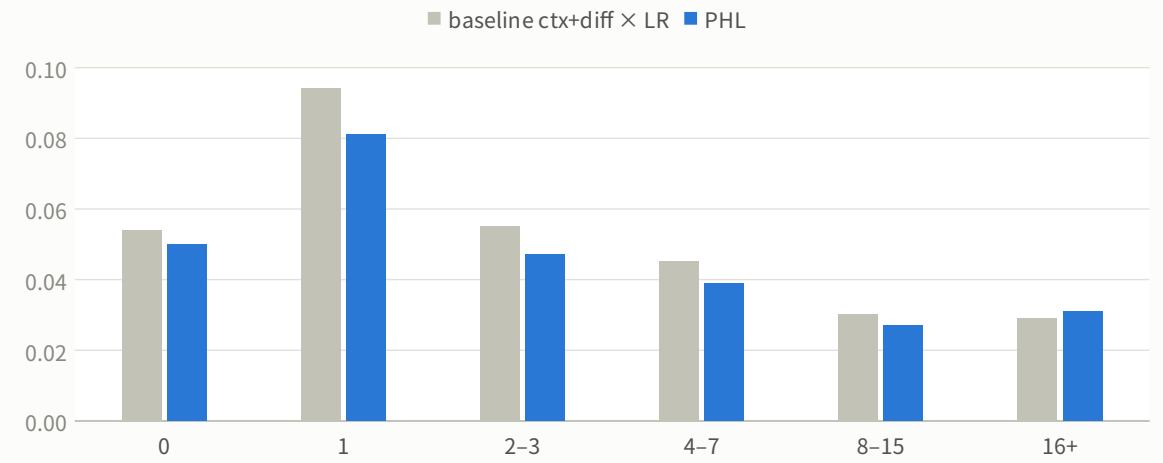
Diagnosis: the warm residual sits on sparse miRNAs, and so does the PHL gain

seed 42, paper 5-fold. Loss = fraction of negatives ranked above each test positive (mean = 1 − AUC); bucketed by miRNA degree in the training support graph.

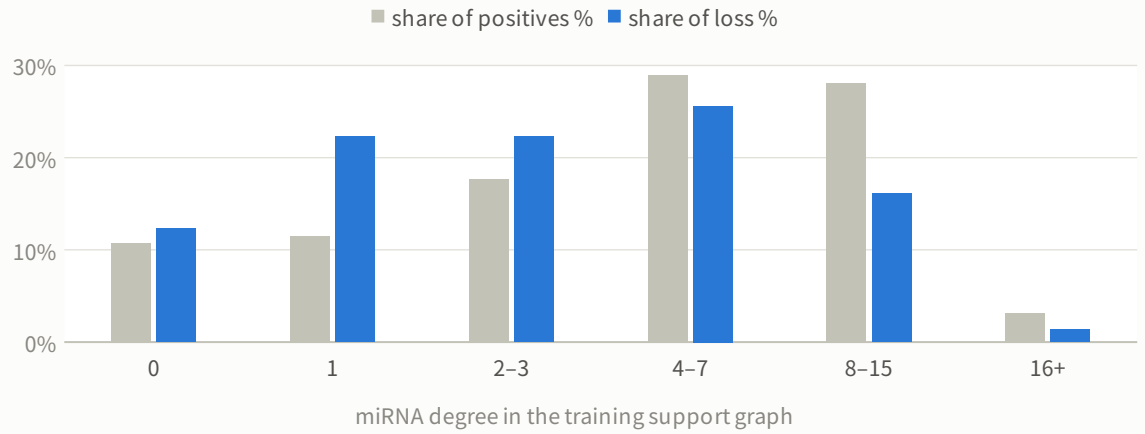
MDR • share of positives vs share of loss — degree ⩽3: 40% of positives, 55% of loss



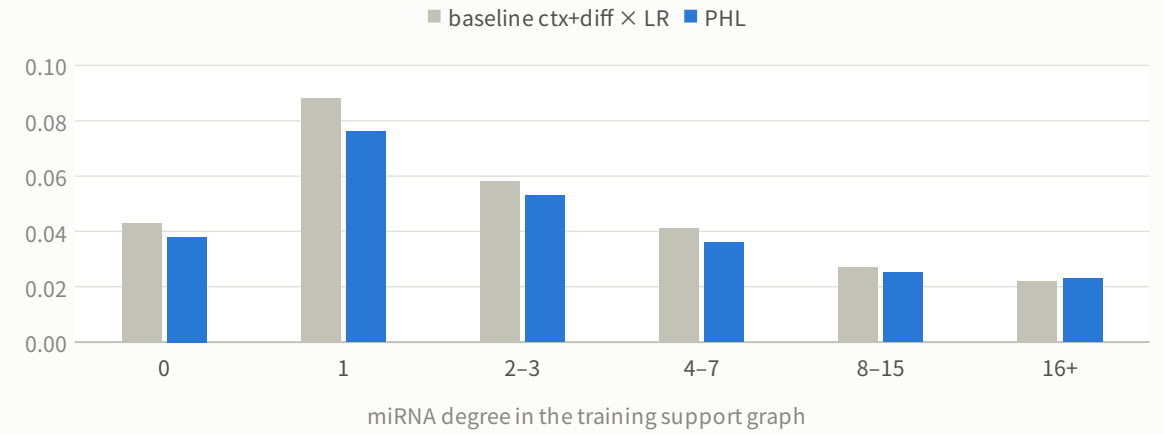
MDR • mean loss per positive, baseline vs PHL — degree 1–7: 13–15% relative drop; 16+ flat



MDS • share of positives vs share of loss — degree ⩽3: 40% of positives, 57% of loss



MDS • mean loss per positive, baseline vs PHL — degree 1–7: 8–14% relative drop; 16+ flat



Numbers from REPORT_WARM_PHL.md § 1.1, § 4.3 (error_analysis.py).